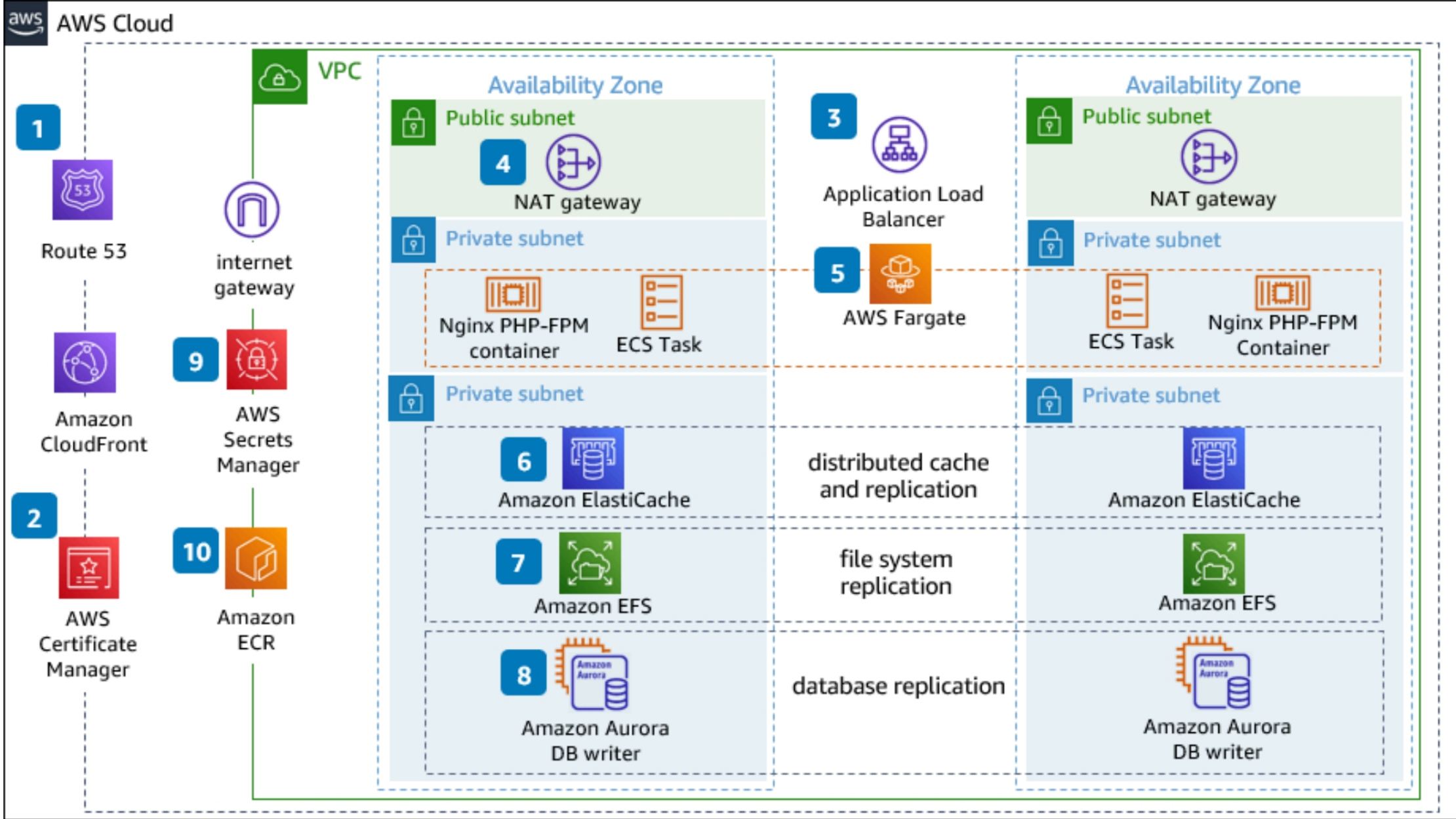


COMPUTER NETWORKING I

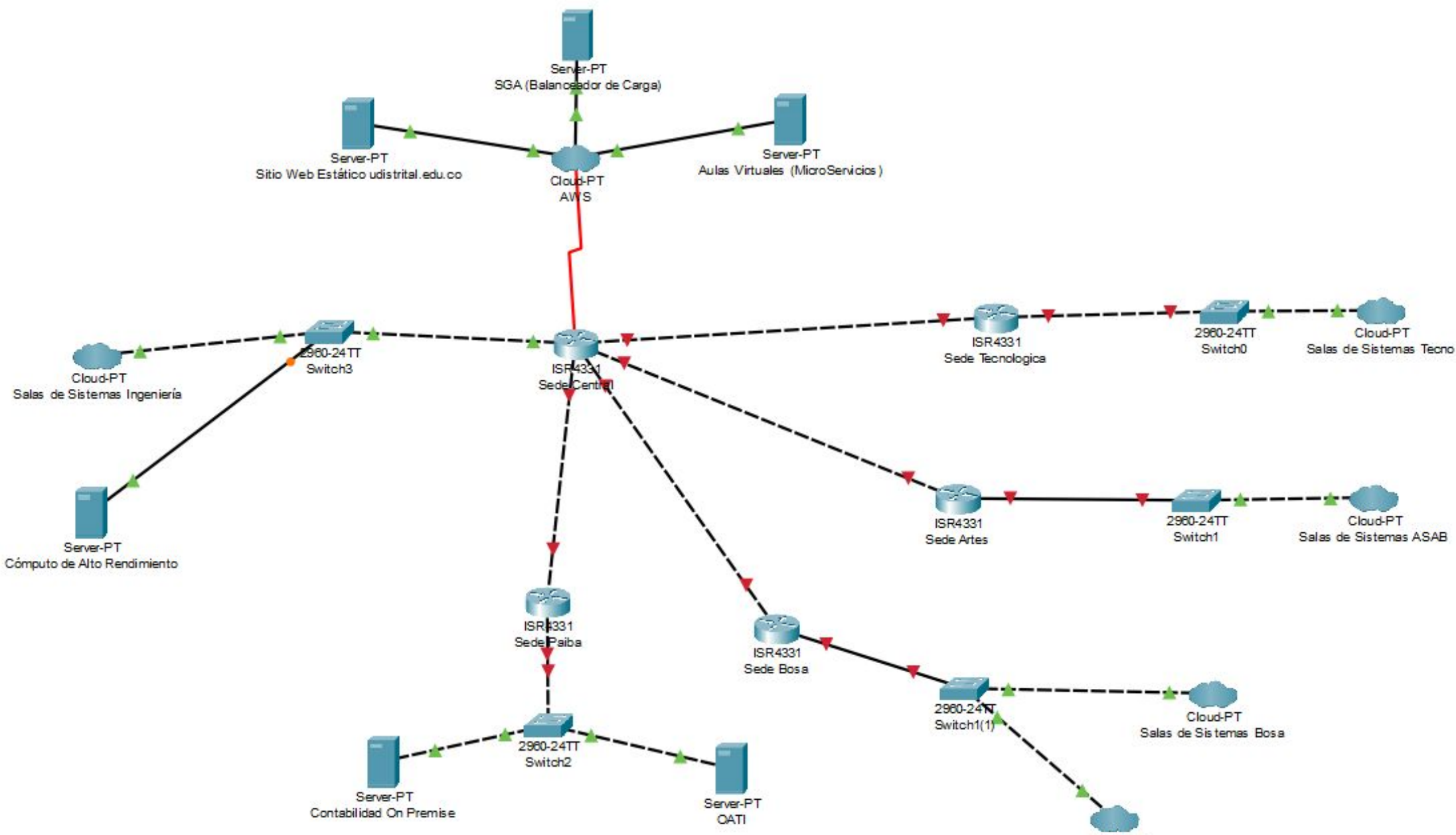
Results

AWS Moodle Architecture



Proposed Architecture

This proposed architecture was made in Cisco Package Tracer.



Budget

This is the first and the expensive budget.

ITEM	QUANTITY	UNITARY PRICE (USD)	TOTAL (USD)
SERVERS			
Blade Servers Chassis	4	15000 USD	60.000 USD
Blade Servers	20	5000 USD	100.000 USD
TOTAL: 160.000 USD			
STORAGE			
NAS System (500 TB)	1	150.000 USD	150.000 USD
TOTAL: 150.000 USD			
NETWORKING			
Core Switches	3	50.000 USD	150.000 USD
Distribution Switches	10	150.00 USD	225.000 USD
Access Points	320	230 USD	73.600 USD
TOTAL: 448.600 USD			
SECURITY			
advanced Firewalls	2	50.000 USD	100.000 USD
IDS/IPS systems	1	50.000 USD	100.000 USD
TOTAL: 200.000 USD			
INFRASTRUCTURE			
Power Distribution	1	300.000 USD	100.000 USD
Cooling System	1	250.000 USD	100.000 USD
Racks and Cabling	1	100.000 USD	100.000 USD
TOTAL: 650.000 USD			
PROFESSIONAL SERVICES			
Design and Distribution	1	500.000 USD	500.000 USD
TOTAL: 500.000 USD			
TOTAL : 2'108.600			

References

[1] Gabriela Martínez Eslava 2020202117, Omar Armando Neira Ordóñez 20192020110, and Edison Mauricio Riaño 20172020113. Computers networking i project, 2025. URL <https://github.com/OmarNeira/ComputersNetworkingIOmarNeira>.

Results

This is the second and the cheaper budget.

Item	Original Cost	Estimated Savings	New Cost
Institutional Email Service	100.000 USD	50.000 USD	50.000 USD
On-Premise Storage (500TB → 250TB)	150.000 USD	75.000 USD	75.000 USD
Networking Equipment (APs Reduced: 230 → 180,)	448.600 USD	75.000 USD	373.600 USD
Security Software & Firewalls (Open Source Alternatives)	150.000 USD	75.000 USD	75.000 USD
Cloud Services (AWS Optimization)	250.000 USD	50.000 USD	200.000 USD
Infrastructure (Energy-Efficient Cooling & Power Optimization)	650.000 USD	95.000 USD	555.000 USD
Professional Services (Delaying Non-Essential Upgrades)	500.000 USD	20.000 USD	480.000 USD
TOTAL	2'108,600 USD	440.000 USD	1.808.600 USD

Conclusion

A high-performance and scalable IT infrastructure is achievable without excessive costs. By migrating non-critical services to the cloud, optimizing resources, and leveraging open-source solutions, operational expenses can be reduced by \$450,000 while ensuring efficiency, security, and reliability. Strategic IT planning reduces costs without compromising performance. Instead of overprovisioning hardware, intelligent architecture and resource allocation optimize expenses while maintaining service quality. Horizontal scaling, cloud optimization, and cost-efficient strategies ensure a future-proof, high-performance, and financially sustainable infrastructure. A robust network directly enhances education by enabling fast, secure, and uninterrupted access to academic resources, virtual learning, and research tools. Networks are the foundation of modern technology, silently powering cloud computing, AI, online education, and global digital communication.